IB MATHEMATICS SL • UNIT 6.01

C Topic 6: AI SL Modeling, Financial Math & Chi-Squared Tests

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 B	Q3 D	Q4 A	Q5 A
Q6 C	Q7 B	Q8 A	Q9 C	Q10 A
Q11 C	Q12 D	Q13 B	Q14 B	Q15 B
Q16 A	Q17 D	Q18 A	Q19 D	Q20 A
Q21 B	Q22 D	Q23 B	Q24 D	Q25 C

Topic 6: AI SL Modeling, Financial Math & Chi-Squared Tests Practice Exercises

1 MCQ

If $P(A) = 0.3$ and $P(B) = 0.5$ and A, B are independent, find $P(A \cup B)$.

Two-Way Contingency Counts

	Outcome +	Outcome –
Group I	42	18
Group II	28	32

A 0.65 ✓

B 0.80

C 0.15

D 0.85

WORKED SOLUTION & VERIFICATION

$P(A \cup B) = 0.3 + 0.5 - 0.3 \times 0.5 = 0.65$ **[M1] [A1]**

2 MCQ

The following bar chart shows the number of students who achieved different grades in a mathematics test.\nThe grades are A, B, C, and D. The data is as follows: Grade A: 14 students, Grade B: 22 students, Grade C: 18 students, Grade D: 26 students.\nWhat is the total number of students who took the test?

A 70

B 80 ✓

C 78

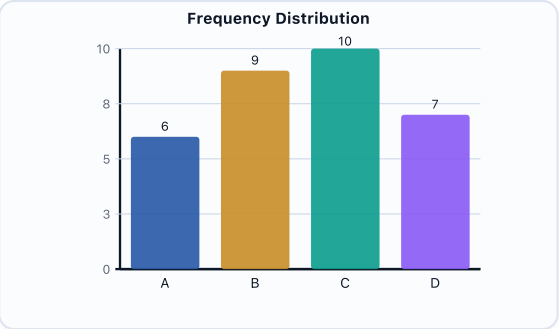
D 60

WORKED SOLUTION & VERIFICATION

To find the total number of students who took the test, we need to sum the number of students who achieved each grade.\nNumber of students with Grade A = 14\nNumber of students with Grade B = 22\nNumber of students with Grade C = 18\nNumber of students with Grade D = 26\nTotal number of students = 14 + 22 + 18 + 26 = 80.\nTherefore, 80 students took the test. **[M1] [A1]**

3 MCQ

The bar chart below shows the number of students who chose each of four different sports as their favorite. What is the total number of students surveyed?



A 50

B 60

C 70

D **80** ✓

WORKED SOLUTION & VERIFICATION

The bar chart shows the following counts for each sport:
Sport 1: 14 students
Sport 2: 22 students
Sport 3: 18 students
Sport 4: 26 students
To find the total number of students surveyed, we sum the number of students for each sport: $14 + 22 + 18 + 26$.
$14 + 22 = 36$
$18 + 26 = 44$
$36 + 44 = 80$.
Therefore, the total number of students surveyed is 80. **[M1] [A1]**

4 MCQ

The histogram below shows the number of hours 100 students spent studying for an exam. The number of students who studied for between 0 and 5 hours is represented by a bar with a height of 30. Determine the number of students who studied for between 5 and 10 hours.

A **20** ✓

B 25

C 30

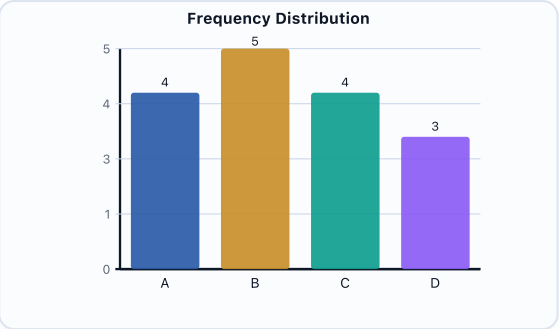
D 35

WORKED SOLUTION & VERIFICATION

The histogram displays the distribution of study hours for 100 students. The total number of students is 100. The question states that the number of students who studied for between 0 and 5 hours is 30. The bar for the '0-5 hours' interval has a height of 30, representing 30 students. The bars for the '5-10 hours' and '10-15 hours' intervals have heights of 20 and 15 respectively, representing 20 and 15 students. Therefore, the number of students who studied for between 5 and 10 hours is 20. **[M1] [A1]**

5 MCQ

The bar chart below shows the number of students who achieved a certain score on a test. What is the total number of students who scored 6 or 8?



- **A** 30 ✓
- **B** 28
- **C** 26
- **D** 32

WORKED SOLUTION & VERIFICATION

From the bar chart, the number of students who scored 6 is 18 and the number of students who scored 8 is 12. The total number of students who scored 6 or 8 is the sum of these two values: $18 + 12 = 30$. **[M1] [A1]**

6 MCQ

The following two-way table shows the results of a survey on favourite sports among a group of students. Calculate the total number of students surveyed.

- **A** 50
- **B** 45
- **C** 60 ✓
- **D** 55

WORKED SOLUTION & VERIFICATION

The total number of students surveyed is the sum of all entries in the two-way table. $\text{Total} = 15 + 10 + 20 + 15 = 60$. **[M1] [A1]**

7 MCQ

Find the mean of the data set: $20, 13, 12, 12, 8$.

- **A** 10
- **B** 13 ✓
- **C** 8
- **D** 11

WORKED SOLUTION & VERIFICATION

Working through this statistics step: $\text{Mean} = \dfrac{20 + 13 + 12 + 12 + 8}{5} = 13$. **[M1] [A1]**

8 MCQ

Find the mean of the data set: $10, 1, 5, 14, 3, 7, 0, 8$.

- **A** 6 ✓
- **B** 4
- **C** 3
- **D** 1

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \dfrac{10+1+5+14+3+7+0+8}{8} = 6$. **[M1] [A1]**

9 MCQ

A particle moves in the xy-plane. Its velocity vector $\mathbf{v}(t)$ at time t is given by $\mathbf{v}(t) = \langle 3t^2 - 6t, e^t - 1 \rangle$, where $t \geq 0$. The initial position of the particle at $t = 0$ is $\mathbf{r}(0) = \langle 5, -2 \rangle$. Find the position vector $\mathbf{r}(t)$ of the particle at time t. Then, determine the value of t for which the particle's speed is minimized and state the minimum speed.

- **A** $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{5}$ at $t = 0$
- **B** $\mathbf{r}(t) = \langle 3t^2 - 6t + 5, e^t - t - 2 \rangle$, minimum speed 1 at $t = 0$
- **C** $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, **minimum speed 0 at $t = 0$** ✓
- **D** $\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle$, minimum speed $\sqrt{2}$ at $t = 1$

WORKED SOLUTION & VERIFICATION

Integrating $\mathbf{v}(t) = \langle 3t^2 - 6t, e^t - 1 \rangle$ gives $\mathbf{r}(t) = \langle t^3 - 3t^2 + C_1, e^t - t + C_2 \rangle$. Using $\mathbf{r}(0) = \langle 5, -2 \rangle$ yields $C_1 = 5$ and $1 + C_2 = -2 \implies C_2 = -3$, so:

$$\mathbf{r}(t) = \langle t^3 - 3t^2 + 5, e^t - t - 3 \rangle.$$

The speed is $|\mathbf{v}(t)| = \sqrt{(3t^2 - 6t)^2 + (e^t - 1)^2} \geq 0$. At $t = 0$, $\mathbf{v}(0) = \langle 0, 0 \rangle$, so the minimum speed is 0 at $t = 0$. **[M1] [A1]**

10 MCQ

In a Voronoi diagram, two sites are located at $A(2, 4)$ and $B(6, 2)$. Which of the following is the equation of the boundary edge lying on the perpendicular bisector of the line segment AB?

A $y = 2x - 5$ ✓

B $y = -\frac{1}{2}x + 5$

C $y = 2x - 8$

D $y = -2x + 11$

WORKED SOLUTION & VERIFICATION

In a Voronoi diagram, the boundary edge between two sites is along the perpendicular bisector of the line segment connecting them.

1. Find the midpoint M of the segment AB:

$$M = (\frac{2+6}{2}, \frac{4+2}{2}) = (4, 3)$$

2. Find the gradient of segment AB:

$$m_{AB} = \frac{2-4}{6-2} = \frac{-2}{4} = -\frac{1}{2}$$

3. The gradient of the perpendicular bisector is the negative reciprocal of m_{AB}:

$$m_{\perp} = -\frac{1}{-\frac{1}{2}} = 2$$

4. Use the point-slope form with midpoint $(4, 3)$ and gradient $m_{\perp} = 2$:

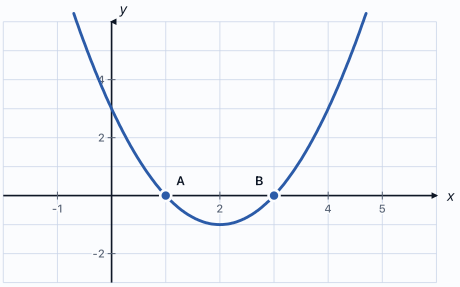
$$y - 3 = 2(x - 4) \implies y = 2x - 8 + 3 \implies y = 2x - 5$$

Common Misconceptions:

- Choice B uses the gradient of AB rather than its negative reciprocal.
- Choice C correctly finds the gradient $m = 2$ but forgets to add the y-coordinate of the midpoint, setting $y = 2(x - 4) = 2x - 8$.
- Choice D uses $m = -2$ instead of $+2$. **[M1] [A1]**

11 MCQ

Consider the quadratic function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. The points where the graph intersects the x-axis are denoted by A and B. What is the distance between points A and B?



A 1

B 3

C 2 ✓

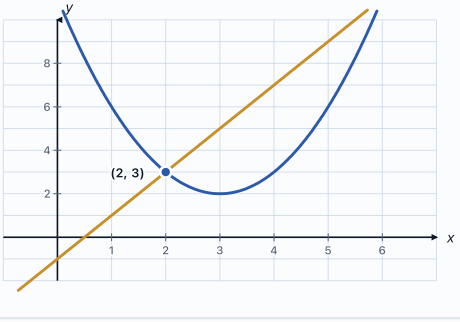
D 4

WORKED SOLUTION & VERIFICATION

The points where the graph intersects the x-axis are the roots of the quadratic equation $f(x) = 0$. We need to solve $x^2 - 4x + 3 = 0$. Factoring the quadratic, we get $(x - 1)(x - 3) = 0$. Thus, the roots are $x = 1$ and $x = 3$. Let point A have coordinates $(1, 0)$ and point B have coordinates $(3, 0)$. The distance between A and B is the absolute difference of their x-coordinates: $|3 - 1| = 2$. **[M1] [A1]**

12 MCQ

Consider the functions $f(x) = x^2 - 6x + 11$ and $g(x) = 2x - 1$. The graphs of these functions intersect at two points. One of the intersection points is $(2, 3)$. Find the x-coordinate of the other intersection point.



A 3

B 4

C 5

D 6 ✓

WORKED SOLUTION & VERIFICATION

To find the intersection points of the two functions, we set $f(x) = g(x)$. $x^2 - 6x + 11 = 2x - 1$

Rearrange the equation to form a quadratic equation:

$x^2 - 6x - 2x + 11 + 1 = 0$

$x^2 - 8x + 12 = 0$

We can factor this quadratic equation. We are looking for two numbers that multiply to 12 and add to -8. These numbers are -6 and -2.

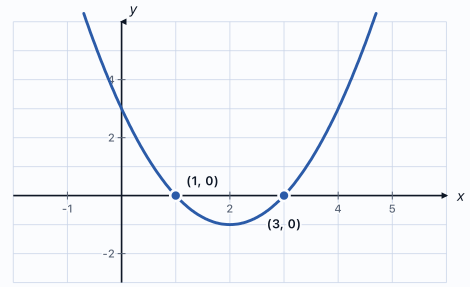
$(x - 2)(x - 6) = 0$

This gives us two possible solutions for x: $x = 2$ or $x = 6$.

We are given that one intersection point has an x-coordinate of 2. Therefore, the x-coordinate of the other intersection point is 6. **[M1] [A1]**

13 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. Find the sum of the x-intercepts of the graph of $y = f(x)$.



A 3

B 4 ✓

C -4

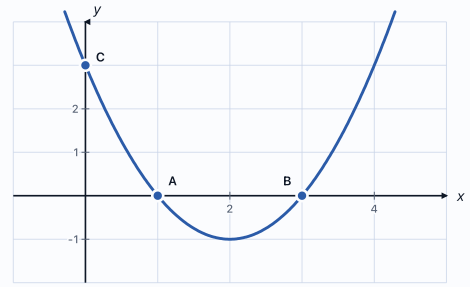
D 1

WORKED SOLUTION & VERIFICATION

The x-intercepts of the graph of $y = f(x)$ occur when $f(x) = 0$. So, we need to solve the quadratic equation $x^2 - 4x + 3 = 0$. This equation can be factored as $(x - 1)(x - 3) = 0$, which gives the x-intercepts as $x = 1$ and $x = 3$. The sum of the x-intercepts is $1 + 3 = 4$. Alternatively, for a quadratic equation of the form $ax^2 + bx + c = 0$, the sum of the roots is given by $-\frac{b}{a}$. In this case, $a = 1$, $b = -4$, and $c = 3$. Thus, the sum of the roots is $-\frac{-4}{1} = 4$. **[M1] [A1]**

14 MCQ

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. Determine the sum of the x-coordinates of A and B.



A -4

B 4 ✓

C 3

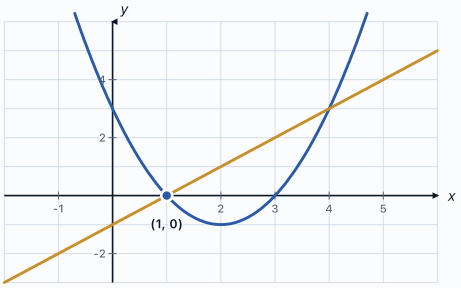
D 1

WORKED SOLUTION & VERIFICATION

For the graph of a quadratic function $f(x) = ax^2 + bx + c$, the sum of the roots (x-intercepts) is given by $-\frac{b}{a}$. In this case, $f(x) = x^2 - 4x + 3$, so $a = 1$ and $b = -4$. Therefore, the sum of the x-coordinates of the points where the graph intersects the x-axis is $-(\frac{-4}{1}) = 4$. Alternatively, we can find the roots by setting $f(x) = 0$: $x^2 - 4x + 3 = 0$. Factoring the quadratic gives $(x - 1)(x - 3) = 0$. The roots are $x = 1$ and $x = 3$. The sum of the roots is $1 + 3 = 4$. **[M1] [A1]**

15 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown. Determine the value of $f(5)$.



A 1

B 8 ✓

C 0

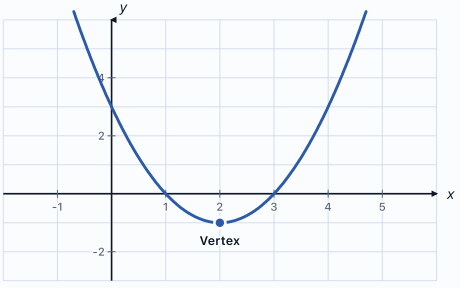
D 18

WORKED SOLUTION & VERIFICATION

To find the value of $f(5)$, we substitute $x = 5$ into the function $f(x) = x^2 - 4x + 3$.
$f(5) = (5)^2 - 4(5) + 3$
$f(5) = 25 - 20 + 3$
$f(5) = 5 + 3$
$f(5) = 8$
Therefore, the value of $f(5)$ is 8. Common errors might arise from miscalculating the square, the subtraction, or the addition. **[M1] [A1]**

16 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.



A 2 ✓

B -2

C 1

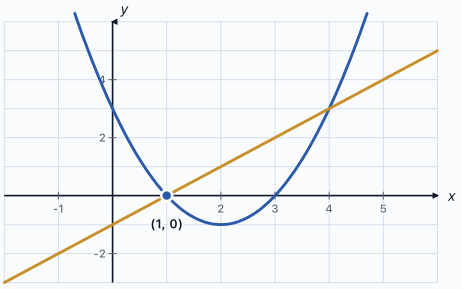
D 3

WORKED SOLUTION & VERIFICATION

The x-coordinate of the vertex of a parabola given by the equation $f(x) = ax^2 + bx + c$ is found using the formula $x = -b/(2a)$. In this case, $a = 1$ and $b = -4$. Therefore, the x-coordinate of the vertex is $x = -(-4)/(2 * 1) = 4/2 = 2$. **[M1] [A1]**

17 MCQ

Consider the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at points A and B. One of these intersection points is $(1, 0)$. Find the coordinates of the other intersection point.



A (3, 2)

B (5, 4)

C (2, 1)

D **(4, 3)** ✓

WORKED SOLUTION & VERIFICATION

To find the intersection points of the two graphs, we set $f(x) = g(x)$.

$x^2 - 4x + 3 = x - 1$

Subtract x from both sides:

$x^2 - 5x + 3 = -1$

Add 1 to both sides:

$x^2 - 5x + 4 = 0$

Factor the quadratic equation:

$(x - 1)(x - 4) = 0$

This gives us two possible x-coordinates for the intersection points: $x = 1$ and $x = 4$.

We are given that one intersection point has an x-coordinate of 1. To find the corresponding y-coordinate, we can substitute $x = 1$ into either $f(x)$ or $g(x)$.

Using $g(x) = x - 1$, we get $g(1) = 1 - 1 = 0$. So the first intersection point is $(1, 0)$.

To find the other intersection point, we use the other x-coordinate, $x = 4$.

Substituting $x = 4$ into $g(x) = x - 1$, we get $g(4) = 4 - 1 = 3$.

Therefore, the other intersection point is $(4, 3)$. **[M1] [A1]**

18 MCQ

$\$1200$ is invested at a simple interest rate of 8% per year for 5 years. Find the interest earned.

A **480.00** ✓

B 470.00

C 490.00

D 500.00

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $I = \frac{Prt}{100} = \frac{(1200)(8)(5)}{100} = 480.00$. **[M1] [A1]**

19 MCQ

Find the midpoint of $A(-1,-1)$ and $B(-7,5)$.

A $(-6,6)$

B $(-3,2)$

C $(-4,3)$

D $(-4,2)$ ✓

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $M=(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2})=(\frac{-1-7}{2},\frac{-1+5}{2})=(-4,2)$. **[M1] [A1]**

20 MCQ

Find the midpoint of $A(1,2)$ and $B(5,10)$.

A $(3,6)$ ✓

B $(3,7)$

C $(4,8)$

D $(4,6)$

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $M=(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2})=(\frac{1+5}{2},\frac{2+10}{2})=(3,6)$. **[M1] [A1]**

21 MCQ

$\$500$ is invested at a simple interest rate of 6% per year for 5 years. Find the interest earned.

A 160.00

B **150.00** ✓

C 170.00

D 140.00

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $I=\frac{Prt}{100}=\frac{(500)(6)(5)}{100}=150.00$. **[M1] [A1]**

22 MCQ

Calculate the midpoint of $A(8,7)$ and $B(2,13)$.

A $(-6,6)$

B $(6,10)$

C $(5,11)$

D $(5,10)$ ✓

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $M=(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2})=(\frac{8+2}{2},\frac{7+13}{2})=(5,10)$. **[M1] [A1]**

23 MCQ

Find the midpoint of $A(1, -6)$ and $B(-9, 2)$.

A $(-4, -1)$

B $(-4, -2)$ ✓

C $(-10, 8)$

D $(-3, -2)$

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}) = (\frac{1 - 9}{2}, \frac{-6 + 2}{2}) = (-4, -2)$. **[M1] [A1]**

24 MCQ

Find the midpoint of $A(-3, -8)$ and $B(-7, -10)$.

A $(-4, -9)$

B $(-4, -2)$

C $(-5, -8)$

D $(-5, -9)$ ✓

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}) = (\frac{-3 - 7}{2}, \frac{-8 - 10}{2}) = (-5, -9)$. **[M1] [A1]**

25 MCQ

$\$2000$ is invested at a simple interest rate of 2% per year for 3 years. Find the interest earned.

A 110.00

B 140.00

C **120.00** ✓

D 130.00

WORKED SOLUTION & VERIFICATION

Working through this modelling step: $I = \frac{Prt}{100} = \frac{(2000)(2)(3)}{100} = 120.00$. **[M1] [A1]**